

Product Document



Application Note

AN001033

AS7343 SMUX Configuration

Filter Channel Mapping by Multiplexing

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1 General Description

The AS7343 integrates a so-called super multiplexer (SMUX). With the SMUX, it is possible to map all available photodiodes to one of the six available light-to-frequency converters (CH0 ADC to CH5 ADC). Every pixel has a multiplexer to map it to one of the engines – this multiplexer can be configured with three bits (0 = pixel disabled or connected to GND; 1 to 6 = ADC 0 to ADC 5). The figures below show the SMUX pixel ID mapping to every individual diode or diodes pair. The “not mentioned” and the grey pixel ID are not used, and will be programmed with “0”. Reading and writing pixel configuration uses the first 10 bytes of the RAM starting at address 00h. For easier usage, the pixel configuration is stored in nibbles within the RAM (4 bits per pixel configuration, MSB not used). It is recommended to write the 20 bytes at once, and configure all the pixels together within one page write command.

Figure 1:
Sensor Array

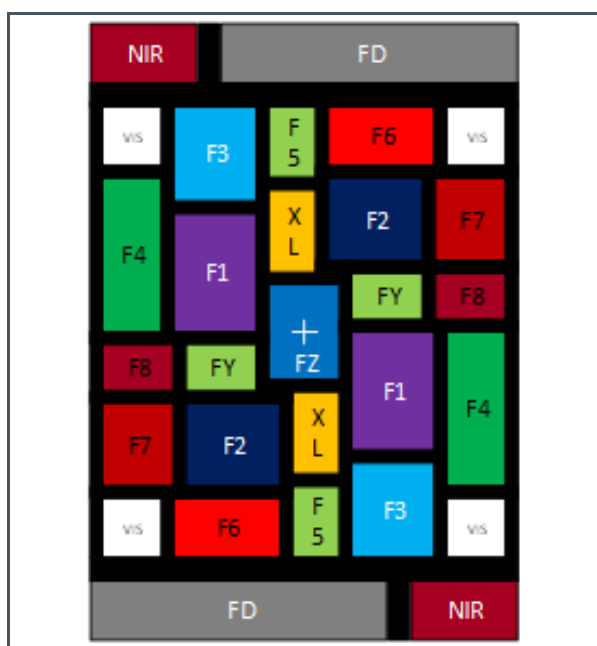


Figure 2:
SMUX Pixel ID Mapping to Diodes

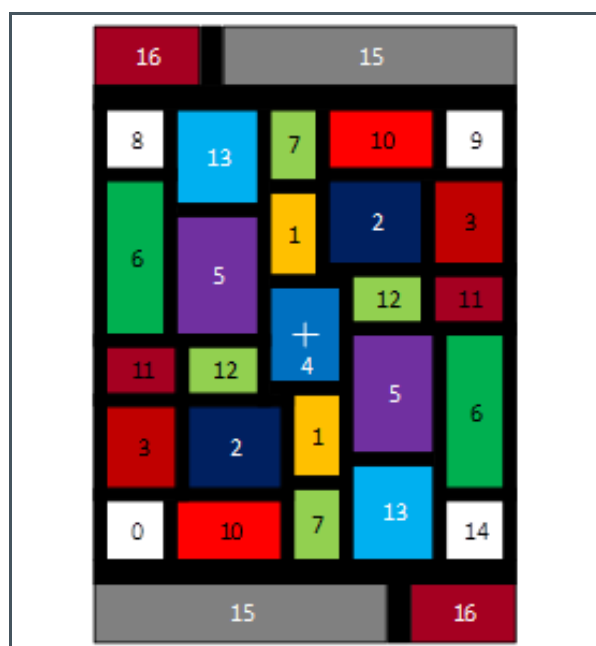


Figure 3:
Chain Map Pixel ID (Line 2) vs Filter ⁽¹⁾ (Line 3)

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
VIS	XL	F2	F7	FZ	F1	F4	F5	VIS	VIS	F6	F8	FY	F3	VIS	FD	NIR	Ext	D	Not used
LB	600	425	690	450	405	515	550	LT	RT	640	745	555	475	RB		855			

(1) VIS = Visible, LB = Left Bottom VIS, LT = Left Top VIS, RT = Right Top, RB = Right Bottom VIS, FD = Flicker Detection, NIR = Near Infrared, D = Dark, C = VIS.

2 SMUX Default Configuration Setting

Besides a default configuration, there are several configuration methods: the RAM method and Direct Chain Configuration. These methods are described in the following chapters of this document.

On the AS7343, there is a default configuration setting for data read-out, in which there is no need to adjust the SMUX in case the default fits the customer requirements.

The default setting is defined as n-integration chains, whereby n can be set in bit `autorun_smux2` (bit 5 and 6 in register CFG20; default 0 – normal six channels) with the settings in Figure 4.

Figure 4:
Settings for Bit `autorun_smux2` to Set Number of Channels for Default SMUX

Value n	Mode	Integration Chain
0 (default)	Normal 6 channels	Integration Chain 1: Z, Y, XL, NIR, VIS ⁽¹⁾ LT+RB, FD
1	Normal 6 channels	Integration Chain 1: Z, Y, XL, NIR, VIS ⁽¹⁾ LT+RB, FD
2	Normal 12 channels	Integration Chain 1: Z, Y, XL, NIR, VIS ⁽¹⁾ LT+RB, FD Integration Chain 2: F2, F3, F4, F6, VIS ⁽¹⁾ LT+RB, FD
3	Normal 18 channels	Integration Chain 1: Z, Y, XL, NIR, VIS ⁽¹⁾ LT+RB, FD Integration Chain 2: F2, F3, F4, F6, VIS ⁽¹⁾ LT+RB, FD Integration Chain 3: F1, F7, F8, F5, VIS ⁽¹⁾ LT+RB, FD

(1) Only two of the four VIS [LT, LB, RT, RB] diodes in the default are read out, to verify a changed light situation within the three Chains.

The data is stored in the data registers accordingly. This means, only six channels are read by using the default 0, or in the case of set $n = 1$. The results of twelve channels from Chains 1 and 2 are in the data registers. Eighteen (18) channels are read out after setting the bit `autorun_smux2` $n = 3$.

3 Specific SMUX Multiplexer Mapping

If the default settings do not meet customer requirements because other channels are needed, or faster measurement time is required, then channel read-out can be configured application-specifically via SMUX configuration.

In the comfortable method (see examples in chapters 3.1 and 3.2) of the “RAM configuration”, the SMUX data from I²C-Addr 0x0 are written into the RAM via the configuration of SMUXEN, and copied with the CMD SMUXEN = 1 by the internal machine into the SMUX chain of the analog part. This method is limited to a configuration with a maximum of six channels (ALS 6) in Chain 1. If more channels/Chains are needed, the following alternative method must be used.

Figure 5:
CHAINSMUX 0x7h Definition

e7h		CHAIN_SMUX		
Bit	Bit Name	Default	Access	Bit Description
7:0	chain_smux	0	R_PUSH	Write: Write to the lower half of the chain shift register, where bits are remapped in the following way: wdata[7:0] = {1'b0, pixel_cfg{i+1}[2:0], 1'b0, pixel_cfg{i}[2:0]} Read: Read from the lower half of the chain return data shift register, where bits are remapped in the following way: rdata[7:0] = {1'b0, channel{i+1}[2:0], 1'b0, channel{i}[2:0]} Note that the pixels are exchanged so it is easier for the ppu to do smux read/write smux configurations with the wanted mapping of the configurations in the RAM.

Figure 6:
CHAINCMD 0x4h Definition

e4h		CHAINCMD			
Bit	Bit Name	Default	Access	Bit Description	
				Setting to non-zero will activate the latch and select the target chain.	
6:4	target_chain	0	R_PUSH	Enumeration	
				Chain	
				0	None
				1	ALSDAC
				2	IDAC
				3	PGDAC
				4	ALSSMUX #1
				5	ALSSMUX #2
3:0	chain_length	0	R_PUSH	6	ALSSMUX #3
				7	None3
				Writing a number (1 to 10) will shift out 1...10 bits from the chain shift register.	

The second method (see the examples in chapters 3.3 and 3.4) is Direct Configuration, where the chain for configuration is addressed directly via CHAINCMD.

Hence, one has direct access to the SMUX-chains 1, 2, and 3 and can change the default configuration for all the ALS 6-12-18 modes (= the maximum number of channels) as desired.

Not depending on the method, the pixel IDs of the diodes must be mapped. The following tables show the principle mapping of the SMUX pixel IDs to the address configuration bit positions in the I²C and RAM address space (for “Chain Map Pixel ID vs. Filter” see Figure 3).

Figure 7:
Mapping of Pixel IDs to RAM Addresses

I ² C Addr.	RAM Addr.	<D7>	<D6>	<D5>	<D4>	<D3>	<D2>	<D1>	<D0>
0x00	0			Pixel ID 1 [6:4]			Pixel ID 0 [2:0]		
0x01				Pixel ID 3 [6:4]			Pixel ID 2 [2:0]		
0x02	1			Pixel ID 5 [6:4]			Pixel ID 4 [2:0]		
0x03				Pixel ID 7 [6:4]			Pixel ID 6 [2:0]		
0x04	2			Pixel ID 9 [6:4]			Pixel ID 8 [2:0]		
0x05				Pixel ID 11 [6:4]			Pixel ID 10 [2:0]		
0x06	3			Pixel ID 13 [6:4]			Pixel ID 12 [2:0]		
0x07				Pixel ID 15 [6:4]			Pixel ID 14 [2:0]		
0x08	4			Pixel ID 17 [6:4]			Pixel ID 16 [2:0]		
0x09				Unused			Pixel ID 18 [2:0]		

Figure 8:
SMUX Multiplexer Mapping

Addr: 0x00 – 0x09		SMUX Multiplexer Mapping		
Bit	Bit Name	Default	Access	Bit Description
7	Reserved	0	RW	Reserved
6:4	Pixel IDx	0	RW	0: Connected to Ground/disabled 1: Pixel connected to ADC0 2: Pixel connected to ADC1 3: Pixel connected to ADC2 4: Pixel connected to ADC3 5: Pixel connected to ADC4 6: Pixel connected to ADC5 (Flicker) 7: Reserved
3	Reserved	0	RW	Reserved
2:0	Pixel IDy	0	RW	0: Connected to Ground/disabled 1: Pixel connected to ADC0

Addr: 0x00 – 0x09		SMUX Multiplexer Mapping		
Bit	Bit Name	Default	Access	Bit Description
				2: Pixel connected to ADC1
				3: Pixel connected to ADC2
				4: Pixel connected to ADC3
				5: Pixel connected to ADC4
				6: Pixel connected to ADC5 (Flicker)
				7: Reserved

3.1 RAM Configuration Readout F1 - F4, VIS, NIR

The following example shows how to map individual PDs to dedicated ADCs using the SMUX. Each box in the chain map example in Figure 9 represents one nibble (4-bit per pixel ID). The number within the box is the value that needs to be programmed to map the pixel to the desired ADC. If no chain is defined via the CHAINCMD register, Chain 1 is always the default configuration target.

- F1 is mapped to ADC0, F2 is mapped to ADC1, F3 is mapped to ADC2, F4 is mapped to ADC3, VIS is mapped to ADC4, and NIR is mapped to ADC5.

Figure 9:
Chain Map Example of Readout F1 - F4, VIS, and NIR (Nibble Sequence Figure 3)

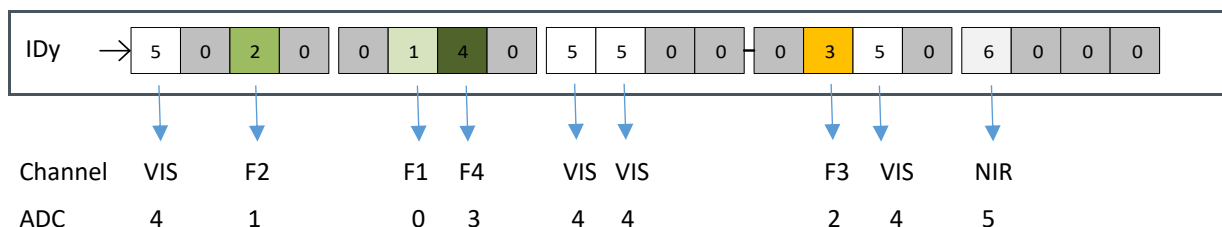


Figure 10:
Configuration Example of Readout F1 - F4, VIS, NIR

Step	I ² C Command	Description
1	I2C_write(0x80, 0x01)	Enable power (set PON = "1") and disable SP_EN (SP_EN="0"). Register: ENABLE/0x80
2	I2C_write(0xCA, 0x10)	Enable special interrupt (SINT_SMUX). As soon as the SMUX command finishes, the interrupt is activated. Register: CFG9/0xCA
3	I2C_write(0xF9, 0x01)	Enable special interrupt SIEN. Register: INTENAB/0xF9

Step	I ² C Command	Description
4	I2C_write(0xF5, 0x10)	Write SMUX configuration from the RAM to set the SMUX chain. Register: CFG6/0xF5
5	I2C_write(0x00, 0x05)	VIS LB connected to ADC4, XL not connected.
6	I2C_write(0x01, 0x02)	F2 connected to ADC1, F7 not connected.
7	I2C_write(0x02, 0x10)	F1 connected to ADC0, FZ not connected.
8	I2C_write(0x03, 0x04)	F4 connected to ADC3, F5 not connected.
9	I2C_write(0x04, 0x55)	VIS LT and VIS RT connected to ADC4.
10	I2C_write(0x05, 0x00)	F6 and F8 not connected.
11	I2C_write(0x06, 0x30)	F3 connected to ADC2, FY not connected.
12	I2C_write(0x07, 0x05)	VIS RB connected to ADC4, FD Flicker not connected.
13	I2C_write(0x08, 0x06)	NIR connected to ADC5, Ext not connected.
14	I2C_write(0x09, 0x00)	D (Dark) not connected.
25	I2C_write(0x80, 0x11)	Start the SMUX command while keeping the power on (SMUXEN = "1" and PON = "1").
26		Wait for interrupt.
27	I2C_write(0x80, 0x00)	Power down (PON = "0").

3.2 RAM Configuration Readout F5 - F8, VIS, NIR

The following example shows how to map individual PDs to dedicated ADCs using the SMUX. If no Chain is defined via the CHAINCMD register, Chain 1 is always the default configuration target.

- F5 is mapped to ADC3, F6 is mapped to ADC0, F7 is mapped to ADC1, F8 is mapped to ADC2, VIS is mapped to ADC4, and NIR is mapped to ADC5.

Figure 11:
Chain Map Example of Readout F5 - F8, VIS, and NIR

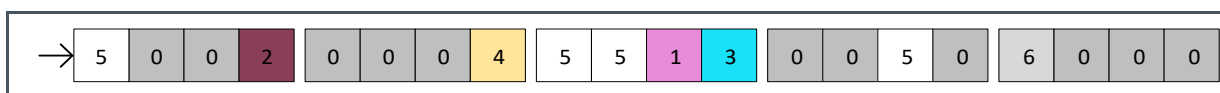


Figure 12:
Configuration Example of Readout F5 - F8, VIS, NIR

Step	I ² C Command	Description
1	I2C_write(0x80, 0x01)	Enable power (set PON = "1"). Register: ENABLE/0x80
2	I2C_write(0xCA, 0x10)	Enable special interrupt (SINT_SMUX). As soon as the SMUX command finishes, the interrupt is activated. Register: CFG9/0xCA
3	I2C_write(0xF9, 0x01)	Enable special interrupt SIEN. Register: INTENAB/0xF9
4	I2C_write(0xF5, 0x10)	Write SMUX configuration from the RAM to set the SMUX chain. Register: CFG6/0xF5
5	I2C_write(0x00, 0x05)	VIS LB connected to ADC4.
6	I2C_write(0x01, 0x20)	F7 connected to ADC1.
7	I2C_write(0x02, 0x00)	
8	I2C_write(0x03, 0x40)	F5 connected to ADC3.
9	I2C_write(0x04, 0x55)	VIS LT VIS RT connected to ADC4.
10	I2C_write(0x05, 0x31)	F6 connected to ADC0. F8 connected to ADC2.
11	I2C_write(0x06, 0x00)	
12	I2C_write(0x07, 0x05)	VIS RB connected to ADC4.
13	I2C_write(0x08, 0x06)	NIR connected to ADC5.
14	I2C_write(0x09, 0x00)	
25	I2C_write(0x80, 0x11)	Start the SMUX command while keeping the power on (SMUXEN = "1" and PON = "1").
26		Wait for interrupt.
27	I2C_write(0x80, 0x00)	Power down (PON = "0")

3.3 Direct Configuration Readout F5 - F8, VIS, NIR

The following example shows how to map individual PDs, plus addressing the Chain n to dedicated ADCs using the SMUX. The (Chain) n is defined as a target for the configuration, by setting 0x46 to register CHAINCMD/0xE4 (alternatively, 0x56 for Chain2 and 0x66 for Chain 3).

- F5 is mapped to ADC3, F6 is mapped to ADC0, F7 is mapped to ADC1, F8 is mapped to ADC2, VIS is mapped to ADC4, and NIR is mapped to ADC5.

Figure 13:
Chain Map Example of Readout F5 - F8, VIS, and NIR

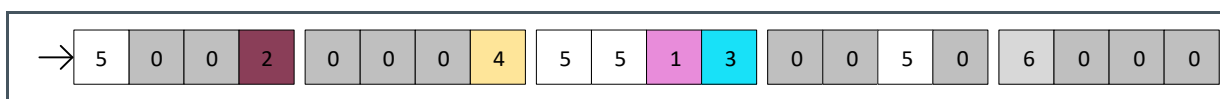


Figure 14:
Direct Configuration Chain1 Example F5-F8, VIS, NIR

Step	I ² C Command	Description
1	I2C_write(0x80, 0x01)	Enable power (set PON = "1"). Register: ENABLE/0x80
2	I2C_write(0xE4, 0x46)	SMUX Chain CMD. Register: CHAINCMD/0xE4
3	I2C_write(0xE7, 0x00)	Write pixel NA 18. Register: CHAIN_SMUX/0xE7
4	I2C_write(0xE4, 0x46)	SMUX Chain CMD. Register: CHAINCMD/0xE4
5	I2C_write(0xE7, 0x06)	Write pixel 17 16 NIR connected to ADC5. Register: CHAIN_SMUX/0xE7
6	I2C_write(0xE4, 0x46)	SMUX Chain CMD. Register: CHAINCMD/0xE4
7	I2C_write(0xE7, 0x05)	Write pixel 15 14 VIS RB connected to ADC4. Register: CHAIN_SMUX/0xE7
8	I2C_write(0xE4, 0x46)	SMUX Chain CMD. Register: CHAINCMD/0xE4
9	I2C_write(0xE7, 0x00)	Write pixel 13 12. Register: CHAIN_SMUX/0xE7
10	I2C_write(0xE4, 0x46)	SMUX Chain CMD. Register: CHAINCMD/0xE4
11	I2C_write(0xE7, 0x31)	Write pixel 11 10 F6 connected to ADC0 F8 connected to ADC2. Register: CHAIN_SMUX/0xE7
12	I2C_write(0xE4, 0x46)	SMUX Chain CMD. Register: CHAINCMD/0xE4 VIS RB connected to ADC4.
13	I2C_write(0xE7, 0x55)	Write pixel 9 8 VIS LT VIS RT connected to ADC4. Register: CHAIN_SMUX/0xE7
14	I2C_write(0xE4, 0x46)	SMUX Chain CMD. Register: CHAINCMD/0xE4
15	I2C_write(0xE7, 0x40)	Write pixel 7 6 F5 connected to ADC3. Register: CHAIN_SMUX/0xE7
16	I2C_write(0xE4, 0x46)	SMUX Chain CMD. Register: CHAINCMD/0xE4
17	I2C_write(0xE7, 0x00)	Write pixel 5 4. Register: CHAIN_SMUX/0xE7
18	I2C_write(0xE4, 0x46)	SMUX Chain CMD. Register: CHAINCMD/0xE4
19	I2C_write(0xE7, 0x20)	Write pixel 3 2 F7 connected to ADC1. Register: CHAIN_SMUX/0xE7

Step	I ² C Command	Description
20	I2C_write(0xE4, 0x46)	SMUX Chain CMD. Register: CHAINCMD/0xE4
21	I2C_write(0xE7, 0x05)	Write pixel 1 0 VISLB connected to ADC4. Register: CHAIN_SMUX/0xE7
22	I2C_write(0x80, 0x00)	Power down (PON = "0").

3.4 Direct Configuration Code Examples for Chain 1 - 3

The following examples show the coded default configuration for Chain 1 – 3, which means for 18 channels (like the settings shown in Figure 4 - version Chain 1 - 3).

Figure 15:
Specification of how to Program Chain 1 – 3

User	0x80	0x01	Set ALSEN=0 LTF ADC off
User	0xe7	0x00	// NC D
User	0xe4	0x46	// smux command chain 1
User	0xe7	0x04	// EXT NIR
User	0xe4	0x46	// smux command chain 1
User	0xe7	0x65	// FLICKER RB
User	0xe4	0x46	// smux command chain 1
User	0xe7	0x02	// F3_475 Y_555
User	0xe4	0x46	// smux command chain 1
User	0xe7	0x00	// F8_745 F6_640
User	0xe4	0x46	// smux command chain 1
User	0xe7	0x05	// RT LT
User	0xe4	0x46	// smux command chain 1
User	0xe7	0x00	// F5_550 F4_515
User	0xe4	0x46	// smux command chain 1
User	0xe7	0x01	// F1_405 Z_450

User	0xe4	0x46	// smux command chain 1
User	0xe7	0x00	// F7_690 F2_425
User	0xe4	0x46	// smux command chain 1
User	0xe7	0x30	// XL_600 LB
User	0xe4	0x46	// smux command chain 1
User	0xe7	0x00	// NC D
User	0xe4	0x56	// smux command chain 2
User	0xe7	0x00	// EXT NIR
User	0xe4	0x56	// smux command chain 2
User	0xe7	0x65	// FLICKER RB
User	0xe4	0x56	// smux command chain 2
User	0xe7	0x20	// F3_475 Y_555
User	0xe4	0x56	// smux command chain 2
User	0xe7	0x04	// F8_745 F6_640
User	0xe4	0x56	// smux command chain 2
User	0xe7	0x05	// RT LT
User	0xe4	0x56	// smux command chain 2
User	0xe7	0x03	// F5_550 F4_515
User	0xe4	0x56	// smux command chain 2
User	0xe7	0x00	// F1_405 Z_450
User	0xe4	0x56	// smux command chain 2
User	0xe7	0x01	// F7_690 F2_425
User	0xe4	0x56	// smux command chain 2
User	0xe7	0x00	// XL_600 LB
User	0xe4	0x56	// smux command chain 2

User	0xe7	0x00	// NC D
User	0xe4	0x66	// smux command chain 3
User	0xe7	0x00	// EXT NIR
User	0xe4	0x66	// smux command chain 3
User	0xe7	0x65	// FLICKER RB
User	0xe4	0x66	// smux command chain 3
User	0xe7	0x00	// F3_475 Y_555
User	0xe4	0x66	// smux command chain 3
User	0xe7	0x30	// F8_745 F6_640
User	0xe4	0x66	// smux command chain 3
User	0xe7	0x05	// RT LT
User	0xe4	0x66	// smux command chain 3
User	0xe7	0x40	// F5_550 F_515
User	0xe4	0x66	// smux command chain 3
User	0xe7	0x10	// F1_405 Z_450
User	0xe4	0x66	// smux command chain 3
User	0xe7	0x20	// F7_690 F2_425
User	0xe4	0x66	// smux command chain 3
User	0xe7	0x00	// XL_600 LB
User	0xe4	0x66	// smux command chain 3

Figure 16:
Example in Python for Three Chains – 18 Channels

```

RA_CFG20 = 0xd6

RA_CHAINCMD = 0xe4

RA_CHAIN_SMUX = 0xe7

# Chain 1 Z_450,Y_555,XL_600,NIR,LT+RB,FLICKER

# Chain 2 F2_425,F3_475,F4_515,F6_640,LT+RB,FLICKER

# Chain 3 F1_405,F7_690,F8_745,F5_550,LT+RB,FLICKER

# NA      D      EXT      NIR      FLICKER      RB  F3_475      Y_555  F8_745  F6_640      RT
LT  F5_550  F4_515  F1_405      Z_450  F7_690  F2_425      XL_600      LB

default_chain1 = [ 0x00, 0x00, 0x00, 0x04, 0x06, 0x05, 0x00, 0x02, 0x00, 0x00,
0x00, 0x05, 0x00, 0x00, 0x01, 0x00, 0x00, 0x03, 0x00 ]

default_chain2 = [ 0x00, 0x00, 0x00, 0x00, 0x06, 0x05, 0x02, 0x00, 0x00, 0x04,
0x00, 0x05, 0x00, 0x03, 0x00, 0x00, 0x00, 0x01, 0x00, 0x00 ]

default_chain3 = [ 0x00, 0x00, 0x00, 0x00, 0x06, 0x05, 0x00, 0x00, 0x03, 0x00,
0x00, 0x05, 0x04, 0x00, 0x01, 0x00, 0x02, 0x00, 0x00, 0x00 ]

# smux configuration without flicker diode based on default

no_fd_chain1  = [ 0x00, 0x00, 0x00, 0x04, 0x00, 0x05, 0x00, 0x02, 0x00, 0x00,
0x00, 0x05, 0x00, 0x00, 0x00, 0x01, 0x00, 0x00, 0x03, 0x00 ]

no_fd_chain2  = [ 0x00, 0x00, 0x00, 0x00, 0x00, 0x05, 0x02, 0x00, 0x00, 0x04,
0x00, 0x05, 0x00, 0x03, 0x00, 0x00, 0x00, 0x01, 0x00, 0x00 ]

no_fd_chain3  = [ 0x00, 0x00, 0x00, 0x00, 0x00, 0x05, 0x00, 0x00, 0x03, 0x00,
0x00, 0x05, 0x04, 0x00, 0x01, 0x00, 0x02, 0x00, 0x00, 0x00 ]

# smux configuration only with flicker diode

flicker_chain = [ 0x00, 0x00, 0x00, 0x00, 0x06, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00 ]

class set_smux:

    def make_chains_default(self):

        for i in range(0,20,2):

```



```

        self.register_write(RA_CHAIN_SMUX, default_chain1[i] << 4 |
default_chain1[i+1])

        self.register_write(RA_CHAINCMD, 0x46)

    for i in range(0,20,2):

        self.register_write(RA_CHAIN_SMUX, default_chain2[i] << 4 |
default_chain2[i+1])

        self.register_write(RA_CHAINCMD, 0x56)

    for i in range(0,20,2):

        self.register_write(RA_CHAIN_SMUX, default_chain3[i] << 4 |
default_chain3[i+1])

        self.register_write(RA_CHAINCMD, 0x66)

def set_smux_default(self):

    self.make_chains_default()

    self.register_write(RA_CFG20, 0x60)

def register_write(self, address, value):

    pass

#end

```

3.5 Chip Library

ams OSRAM provides a chip library for the AS7343 upon request. This library provides low-level access to the sensor as well as basic configuration possibilities including SMUX settings. However, the firmware for a sensor is always dependent on the system configuration used, i.e. the hardware and software of the sensor and the host computer. Therefore, the ChipLib is generic and considers exemplary hardware and software configurations.

4 Revision Information

Changes from previous version to current revision v3-00	Page
Figure 3 filled with values according to the AS7343 datasheet.	4
Xk changed to XL in Figure 4.	5
New figures for CHAINSMUX (Figure 5) and CHAINCMD (Figure 6).	6
Channel order correction in chains 2 and 3: F5 changed to F6 in step 10 of Figure 12, F6 changed to F7 in step 6 of Figure 12, F7 changed to F8 in step 10 of Figure 12, F8 changed to F5 in step 8 of Figure 12.	10
Changes to ADC mapping: F5 mapping changed from ADC0 to ADC3, F6 mapping changed from ADC1 to ADC0, F7 mapping changed from ADC2 to ADC1, F8 mapping changed from ADC3 to ADC2.	10
Corrections to Chain 1 table example: F5 changed to F6 and F7 changed to F8 in step 11 of Figure 14, F8 changed to F5 in step 15 of Figure 14, F6 changed to F7 in step 19 of Figure 14.	11
Changes to the values and comments in Figure 15.	12
Changes to the values and comments in Figure 16	15

- Page and figure numbers for the previous version may differ from page and figure numbers in the current revision.
- Correction of typographical errors is not explicitly mentioned.

5 Legal Information

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